



Lakhshya JEE 2.0 (2024)

Determinant

DPP-04

1. If $a > 0, b > 0, c > 0$ are respectively the $p^{\text{th}}, q^{\text{th}}, r^{\text{th}}$ terms of a G.P., then the value of the determinant

$$\begin{vmatrix} \log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1 \end{vmatrix} \text{ is}$$

- (1) 1 (2) -1
(3) $\frac{abc}{pqr}$ (4) 0

2. If $I_n = \begin{vmatrix} 1 & k & k \\ 2n & k^2 + k + 1 & k^2 + k \\ 2n-1 & k^2 & k^2 + k + 1 \end{vmatrix}$ and

$$\sum_{n=1}^k I_n = 72, \text{ then } k =$$

- (1) 8 (2) 9
(3) 6 (4) none of these

3. If $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix} = x + iy$, then

- (1) $x = 3, y = 1$ (2) $x = 1, y = 3$
(3) $x = 0, y = 3$ (4) $x = 0, y = 0$

4. Let $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, then the value of the

$$\text{determinant } \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 - \omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix} \text{ is:}$$

- (1) 3ω (2) $3\omega(\omega - 1)$
(3) $3\omega^2$ (4) $3\omega(1 - \omega)$

5. If $\Delta_r = \begin{vmatrix} 1 & r & 2^r \\ 2 & n & n^2 \\ n & \frac{n(n+1)}{2} & 2^{n+1} \end{vmatrix}$, then the value of

$$\sum_{r=1}^n \Delta_r \text{ is}$$

- (1) n (2) $2n$
(3) $-2n$ (4) n^2

6. The system of equation,
 $\lambda x + y + z = 0,$
 $-x + \lambda y + z = 0,$
 $-x - y + \lambda z = 0$
will have a non-zero solution if real values of λ are given by:

- (1) 5 (2) -1
(3) 1 (4) 0

7. The value of the determinant

$$\begin{vmatrix} \sin^2\left(x + \frac{3\pi}{2}\right) & \sin^2\left(x + \frac{5\pi}{2}\right) & \sin^2\left(x + \frac{7\pi}{2}\right) \\ \sin\left(x + \frac{3\pi}{2}\right) & \sin\left(x + \frac{5\pi}{2}\right) & \sin\left(x + \frac{7\pi}{2}\right) \\ \sin\left(x - \frac{3\pi}{2}\right) & \sin\left(x - \frac{5\pi}{2}\right) & \sin\left(x - \frac{7\pi}{2}\right) \end{vmatrix} \text{ is}$$

- (1) 1 (2) 2
(3) 3 (4) 0

8. Let $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, then the value of the

$$\text{determinant } \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 - \omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix}, \text{ is}$$

- (1) 3ω (2) $3\omega(\omega - 1)$
(3) $3\omega^2$ (4) $3\omega(1 - \omega)$



9. The value of the determinant $\begin{vmatrix} bc & ca & ab \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$, where

a, b, c are the p^{th} , q^{th} and r^{th} terms of a H.P., is

- (1) $p + q + r$ (2) $(a + b + c)$
(3) 1 (4) None of these

10. If a, b, c are in AP, then the value of

$$\begin{vmatrix} x+2 & x+3 & x+a \\ x+4 & x+5 & x+b \\ x+6 & x+7 & x+c \end{vmatrix} \text{ is}$$

- (1) $x - (a + b + c)$ (2) $9x^2 + a + b + c$
(3) $a + b + c$ (4) 0

11. If w is a complex cube root of unity.

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = - (a + b + c) (a + bk + ck^2)$$

($a + bk^2 + ck$), then k equals

- (1) 1 (2) -1
(3) ω (4) $-\omega$

12. If $\Delta_r = \begin{vmatrix} 2^{r-1} & 2 \cdot 3^{r-1} & 4 \cdot 5^{r-1} \\ x & y & z \\ 2^n - 1 & 3^n - 1 & 5^n - 1 \end{vmatrix}$, then $\sum_{r=1}^n \Delta_r$ is equal

to

- (1) xyz (2) 1
(3) -1 (4) 0

13. If $\Delta_r = \begin{vmatrix} r & x & \frac{n(n+1)}{2} \\ 2r-1 & y & n^2 \\ 3r-2 & z & \frac{n(3n-1)}{2} \end{vmatrix}$, then $\sum_{r=1}^n \Delta_r$ is equal

to

- (1) xyz (2) $nxyz$
(3) 0 (4) none of these

14. The digits A, B, C are such that the three digit numbers $A88, 6B8, 86C$ are divisible by 72, then

the determinant $\begin{vmatrix} A & 6 & 8 \\ 8 & B & 6 \\ 8 & 8 & C \end{vmatrix}$ is divisible by

- (1) 72 (2) 144
(3) 288 (4) 1216

15. Let $P = [a_{ij}]$ be a 3×3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j}a_{ij}$ for $1 < i, j \leq 3$. If the determinant of P is 2, then the determinant of the matrix Q is

- (1) 2^{10} (2) 2^{11}
(3) 2^{12} (4) 2^{13}

16. If $\Delta = \begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$, then the $\frac{1}{1000}$

[maximum value of Δ - minimum value of Δ]³ is equal to ____.

- (1) 1 (2) -1
(3) 0 (4) 2



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (4)
2. (1)
3. (4)
4. (2)
5. (3)
6. (4)
7. (4)
8. (2)

9. (1)
10. (4)
11. (3)
12. (4)
13. (3)
14. (1, 2, 3)
15. (4)
16. (1)



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